OPENVALS CASE STUDY 001

Evaluating the Evaluator — LLM-as-a-Judge Reliability

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Engineering AI Trust. Measuring Enterprise Value.

Executive summary

This research package demonstrates how an AI judging process can be distorted by response order and verbosity. It uses a deterministic synthetic simulation, not a live LLM or production OpenVals engine.

Design and mathematical model

Six synthetic IT-support cases contain expert-preferred safe responses and nonpreferred unsafe alternatives; four are marked critical. The simulated score is S(r,p) = Q(r) + alpha * I(first position) + beta * word_count(r). Both answer orders are tested.

Synthetic experiment outputs (n = 6)

Configuration	AB accuracy	Order flips	Critical false approvals
unbiased oracle	100%	0%	0%
position bias only	50%	100%	50%
verbosity bias only	0%	0%	100%
combined bias	50%	83%	50%

Evidence notice: All values are synthetic script outputs, not real-model findings.

What the metrics mean

AB accuracy measures agreement with authored reference labels. Order-swap inconsistency measures changes in winner identity after swapping positions. Critical incorrect approval rate measures nonpreferred AB selections among the four critical cases.

Limitations

The data, labels, and judge behavior are constructed. The judge receives oracle quality directly. Results cannot establish real-world prevalence, model quality, OpenVals performance, or enterprise ROI.

Next step: enterprise validation

Use a representative permissioned dataset, blinded independent expert labels, prespecified rubrics, versioned judges, repeated balanced comparisons, uncertainty estimates and a validated OpenVals adapter. Compare human-review burden and false approvals with a measured baseline.

Reproduction

Run: python3 case-studies/001-llm-judge-reliability/experiment.py --output reports/results.json ; then python3 -m unittest discover -s tests -v. No external dependencies.